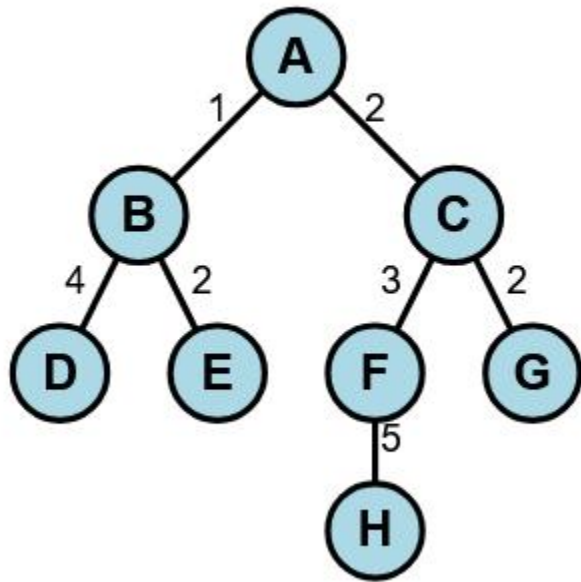




Node	$h(n)$
A	10
B	8
C	5
D	7
E	6
F	3
G	6
H	0

Start Node: A, Goal Node: H

Find the best path using the following BFS, DFS, and A*



Index of 8 Queens

6	1	7	3	6	1	4	0
3	4	0	6	6	1	2	3
0	3	7	0	1	1	4	0

- 1) Calculate Fitness of everyone.
- 2) What is the best population to choose more than once and why?



4-Queens Problem

Place 4 queens on a 4x4 chessboard such that:

- No two queens attack each other
- Each queen must be in a different row and column

Goal:

Formulate the **problem as a CSP** and find a **valid assignment using Backtracking**.

Explain **Variables, Domains, and Constraints**.

University Exam Scheduling

A university needs to schedule final exams for 3 courses: AI, Compiler and Security.

Available Time Slots:

- Slot 1: Monday 9AM
- Slot 2: Monday 1PM
- Slot 3: Tuesday 9AM

Constraints:

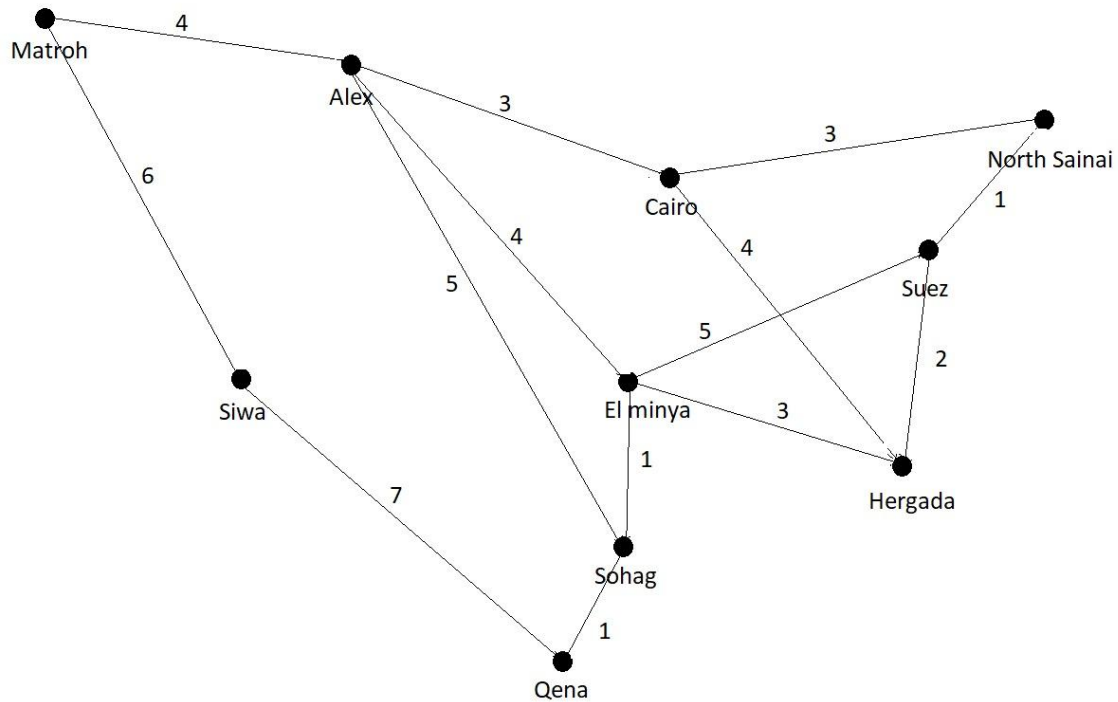
- A student is taking both AI and Compiler $\rightarrow$ AI $\neq$ Compiler
- Compiler professor is only available on Monday $\rightarrow$ Compiler $\in$ {Slot1, Slot2}
- Only one exam hall is available $\rightarrow$ No two exams can be scheduled at the same time

? Goal:

Assign **a time slot to each course satisfying** all constraints.

Explain **Variables, Domains, Constraints** and **Backtracking** steps.

Q



<i>Node Names</i>	<i>Heuristics Cost</i>
Matroh	6
Alex	3
Siwa	4
Cairo	7
El minya	2
Sohag	8
Qena	1
Sainai	5
Suez	3
Hergada	6

- Find the best path using the following BFS, DFS, UCS, and A*:
 - Considering 'Matroh' is your start and 'Suez' is your goal.
 - Considering 'Qena' is your start and 'Cairo' is your goal.

Q: Using detailed steps (Pseudocode | Algorithm) Paraphrase how can you use genetic algorithm in solving sudoku game.